Counting, Carrying, and Circuits

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1. Carrying is the same in every base. A base-b system has b symbols; the biggest is $b-1$. When a place runs out of symbols, write 0 there and add 1 to the next place.

	biggest digit +1	**two biggest +1**	**place values**
base 10	$9+1=10$	$99+1=100$	$1, 10, 100$
base 8	$7+1=10$	$77+1=100$	$1, 8, 64$
base 16	$F+1=10$	$FF+1=100$	$1, 16, 256$
base 2	$1+1=10$	$11+1=100$	$1, 2, 4, 8$

"10" always means *one group of b, none left over.* The marks could be anything: circle, star, face, fish.

Converting. *Binary → decimal:* write the place values above the bits, add the places that hold a 1. `1000101`: $64+4+1=69$. `10011101`: $128+16+8+4+1=157$. Seven digits means the top place is $2^6=64$ (places count from 0); leading zeros change nothing. *Decimal → binary,* two ways. (a) Biggest power that fits: $69=64+4+1$, so `1000101`. (b) Divide by 2 and read the remainders *upward*: $69\to 34\,r1,\ 34\to 17\,r0,\ 17\to 8\,r1,\ 8\to 4\,r0,\ 4\to 2\,r0,\ 2\to 1\,r0,\ 1\to 0\,r1$, which reads `1000101`. Both give the same answer.

2. Add one column at a time, right to left, then move sideways. Line the digits up. Write any carry above the next column.

place	100s	10s	1s
carry	1		
		5	2
+		5	3
	1	0	5

Ones: $2+3=5$.
Tens: $5+5=10$. Write 0, **carry 1**.
100s: nothing to add, so the carry drops down: 1.
Answer: 105.

Binary uses the identical steps, with a smaller table: $0+0=0,\ 0+1=1,\ 1+0=1,\ 1+1=$ `10` (write 0, carry 1). Each example also shows the numbers converted to decimal, so you can check that the math works the same way *both directions*: binary → decimal, and decimal → binary.

Short $(3+1=4)$

place	4s	2s	1s
carry	1	1	
		1	1
+			1
	1	0	0

`11` $=2+1=3$; `1` $=1=1$; sum $3+1=4=$ `100`

Medium $(11+6=17)$

place	16s	8s	4s	2s	1s
carry	1	1	1	0	
		1	0	1	1
+		0	1	1	0
	1	0	0	0	1

1s: $1+0=1$. **2s:** $1+1=10$, write 0, carry 1.
4s: $0+1+1=10$, write 0, carry 1.
8s: $1+0+1=10$, write 0, carry 1.
16s: the carry drops down: 1.

`1011` $=8+2+1=11$; `0110` $=4+2=6$; sum $11+6=17=16+1=$ `10001`

Long $(183+109=292)$

place	256s	128s	64s	32s	16s	8s	4s	2s	1s
carry	1	1	1	1	1	1	1	1	
		1	0	1	1	0	1	1	1
+		0	1	1	0	1	1	0	1
	1	0	0	1	0	0	1	0	0

`10110111` $=128+32+16+4+2+1=183$; `01101101` $=64+32+8+4+1=109$; sum $183+109=292=256+32+4=$ `100100100`

3. Multiply the same way in every base: one partial product per digit of the bottom number, each shifted left one place, then add the columns. In binary a partial product is either a copy of the top number or all zeros.

Decimal first $(23 \times 14 = 322)$

$$\begin{array}{cccc} & & 2 & 3 \\ \times & & 1 & 4 \\ \hline & & 9 & 2 \\ & 2 & 3 & 0 \\ \hline & 3 & 2 & 2 \end{array}$$

$23 \times 4 = 92$ (the ones digit of the bottom number)
$23 \times 10 = 230$ (the tens digit: shifted left one place)
$92 + 230 = 322$. Add the columns, right to left, with carries.

Binary example 1 $(5 \times 3 = 15)$

$$\begin{array}{cccc} & 1 & 0 & 1 \\ \times & & 1 & 1 \\ \hline & 1 & 0 & 1 \\ 1 & 0 & 1 & \\ \hline 1 & 1 & 1 & 1 \end{array}$$

$\texttt{101} = 5$; $\texttt{11} = 3$; product $5 \times 3 = 15 = 8 + 4 + 2 + 1 = \texttt{1111}$

Binary example 2 $(6 \times 5 = 30)$

$$\begin{array}{ccccc} & & 1 & 1 & 0 \\ \times & & 1 & 0 & 1 \\ \hline & & 1 & 1 & 0 \\ & 0 & 0 & 0 & \\ 1 & 1 & 0 & & \\ \hline 1 & 1 & 1 & 1 & 0 \end{array}$$

$\texttt{110} = 6$; $\texttt{101} = 5$; product $6 \times 5 = 30 = 16 + 8 + 4 + 2 = \texttt{11110}$

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4. Boolean algebra. Values 0 and 1. **AND** $A \cdot B$ is 1 only if both are 1. **OR** $A + B$ is 1 if at least one is 1. **NOT** $\overline{A}$ flips. **XOR** $A \oplus B$ is 1 when the inputs differ.

Careful: in Boolean algebra $1 + 1 = 1$ (OR). In binary arithmetic $1 + 1 = 10$ (add). Always say which system you are in.

5. Half adder (adds two bits). Sum $S = A \oplus B$. Carry $C = A \cdot B$. Read CS as a two-bit number: it equals $A + B$.

A	B	C	S	$A + B$ in binary
0	0	0	0	00
0	1	0	1	01
1	0	0	1	01
1	1	1	0	10

6. Full adder (adds A, B, and a carry-in). $S = A \oplus B \oplus C_{in}$ $\quad C_{out} = A \cdot B + C_{in} \cdot (A \oplus B)$.

A	B	C_{in}	C_{out}	S	$A + B + C_{in}$
0	0	0	0	0	00
0	0	1	0	1	01
0	1	0	0	1	01
0	1	1	1	0	10
1	0	0	0	1	01
1	0	1	1	0	10
1	1	0	1	0	10
1	1	1	1	1	11

All eight input rows are listed, so the table is a **proof**, not just an example.

7. Chaining adders. One full adder per bit, each carry feeding the next: a **ripple-carry adder**. The carry travels through every stage, so about $2n$ gate delays for n bits (8 bits: 16; 16 bits: 32). **Carry lookahead** computes the carries directly from *generate* $G_i = A_i B_i$ and *propagate* $P_i = A_i \oplus B_i$: $C_1 = G_0 + P_0 C_0$, $C_2 = G_1 + P_1 G_0 + P_1 P_0 C_0$. A 16-bit lookahead adder finishes in about 8 gate delays instead of 32, for more hardware.